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collected data is saved locally in a .csv file, enabling the application to visualise usage history, generate electricity bill predictions using

a machine learning model, and display everything through an intuitive and responsive dashboard.

The UI provides three options:

- **Select month to view usage.** Selecting a specific month and year displays energy usage and previous bills for that period
- **Real-time energy stats.** Displays the total power consumed from the start of the day
- **Bill prediction options.** Predicts the next month's electricity bill and, if required, estimates bills for the remaining months of the year

Machine learning model: Predicting your bills

For predictive analysis, the system

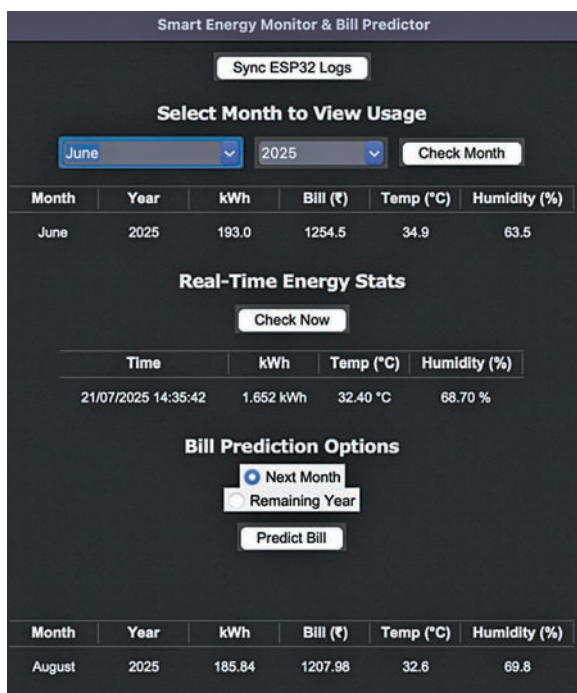


Fig. 6: Sample of the UI

```
def fetch_realtime_data():
    global daily_energy_total, daily_temp_readings, daily_hum_readings, last_logged_date
    try:
        response = requests.get(ESP32_URL, timeout=5)
        if response.status_code == 200:
            raw = response.text.strip()

            # Log format: "13/07/2025,2.56,32.4,65.0"
            if ',' in raw and raw.count(',') == 3:
                date, kwh, temp, hum = raw.split(',')

            # Live format:
            # Live Energy: 1.2345 kWh
            # Temp: 32.55 °C
            # Humidity: 67.33 %
            elif "Live Energy" in raw:
                lines = raw.split('\n')
                if len(lines) < 3:
                    raise ValueError("Incomplete ESP32 live data format")

                kwh = float(lines[0].split(':')[1].strip().split()[0])
                temp = float(lines[1].split(':')[1].strip().split()[0])
                hum = float(lines[2].split(':')[1].strip().split()[0])
                date = datetime.now().strftime("%d/%m/%Y")

            else:
                raise ValueError("Unknown ESP32 response format")

            # Store for monthly log
            if date != last_logged_date:
                last_logged_date = date
                daily_energy_total += float(kwh)
                daily_temp_readings.append(float(temp))
                daily_hum_readings.append(float(hum))
```

Fig. 7: Snippet of Tkinter code